Chaeun Lee

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RESEARCH PROFILE

I want to understand how an AI system can turn experience into a theory of the world. Rather than memorizing tasks or merely predicting the next state, I am interested in world models that form generalizable insights from experience, uncover the mechanisms behind them, and recombine those insights to solve unfamiliar reasoning problems. I am particularly interested in how such models can revise their own explanations through interaction and support embodied agents that must reason and act under uncertainty. I also want to study how model architectures and their execution can be designed together so that increasingly capable reasoning models remain efficient under real compute, memory, and data-movement constraints.

EDUCATION

Sungkyunkwan University (SKKU) **Mar 2023 - Feb 2027**

B.S. in Systems Management Engineering | Second Major: Advanced Semiconductor Engineering *Expected Feb 2027*

- Overall GPA: 4.02 / 4.5 (95.2/100) | Major GPA: 4.1 / 4.5
- Academic Excellence Scholarship: Fall 2024, Spring 2025
- Relevant Coursework: Parallel Computer Architecture & Programming; AI System Semiconductor Design; AI System Design

PUBLICATION

S. Choi, W. Choi, S. Chun, D. Lee, J. Kim, **C. Lee**, H. Woo. "Efficient Skill Grounding via Code Refactoring with Small Language Models." *International Conference on Machine Learning (ICML), PMLR 306, 2026.* [arXiv:2606.07999]

RESEARCH EXPERIENCE

Machine Learning and Mind Lab (MLML), KAIST **Jun 2026 - Aug 2026**

Research Intern | Advisor: Prof. Sungjin Ahn

- Reproduced the Generative Recursive Model (GRAM) on N-Queens, Graph Coloring, ARC-AGI, and Sudoku, including data generation, training, and evaluation.
- Decomposed reasoning performance into solution generation and selection, finding that the main bottleneck came from failing to generate correct reasoning trajectories rather than selecting among generated candidates.
- Investigated how stochastic latent noise affects reasoning accuracy and solution coverage through controlled experiments, including inference-time noise scaling.
- Explored alternative reasoning and search mechanisms using concepts from SMC, flow-based generation, and fixed-point reasoning.

CSI-Agent Lab, Sungkyunkwan University **Oct 2025 - May 2026**

Undergraduate Research Intern | Advisor: Prof. Honguk Woo

- Implemented RepairAgent, the main baseline of our ICML 2026 work, for sLM-based embodied robot control.
- Identified embodied failures where code terminates successfully but the robot fails to reach the intended task state, separating execution validity from actual goal achievement.
- Implemented recovery logic that decides whether to repair failed code, continue from the current world state, or regenerate the plan based on the failure and environment state.
- Structured object/gripper states, previous actions, task constraints, and failure signals for model feedback, while handling syntactic errors through AST-based post-processing.
- Evaluated the system in RLBench/CoppeliaSim and Genesis using task success, execution failures, token usage, interruptions, and waiting time.

RESEARCH EXPERIENCE

System Intelligence Lab, KAIST **Jun 2025 - Aug 2025**

Undergraduate Research Intern | Advisor: Prof. Jinkyoo Park

- Studied world models including DreamerV3 and V-JEPA, model-based reinforcement learning, and score-based diffusion through weekly paper discussions and lectures.
- Conducted Diffusion Posterior Sampling experiments for denoising and inpainting inverse problems and presented the results in a lab seminar.

SELECTED AI SYSTEMS & PROJECTS

NPU Workload Profiling & HW/SW Optimization **Sep 2026 - Present**

BOS Semiconductor / SKKU

- Profiled VLA inference workloads on a BOS Semiconductor NPU, focusing on attention, tensor layout, memory movement, and core utilization.
- Identified repeated DRAM traffic during Interleaved-to-Sharded layout conversions as a major data-movement bottleneck.
- Moved intermediate activations from DRAM to L1 SRAM, reducing layout-conversion overhead by 77.6%.
- Extended the analysis to tensor sharding, resharding, core utilization, and memory placement across VLA workloads.

AI-Based Bad Pixel Detection & Restoration

STOB League 2025

- Developed a 3-head U-Net-based model for joint bad-pixel detection, correction, and demosaicing in CMOS image sensors.
- Applied INT4 quantization (GPTQ) and sparsity pruning (SparseGPT) for NPU-oriented deployment.
- Achieved 11.43x model compression, 87.5% lower memory usage, and 63.6% lower inference latency, while limiting PSNR degradation to approximately 1 dB.

SELECTED COURSE PROJECTS

CUDA Kernel Optimization

Parallel Computer Architecture & Programming

- Implemented tiled GEMM using 32x32 shared-memory tiles and achieved up to 1.96x speedup over the global-memory baseline by increasing data reuse.

AI Accelerator RTL Implementation

AI System Semiconductor Design

- Converted a PyTorch MNIST DNN into 16-bit fixed-point SystemVerilog RTL, including memory placement and inter-layer data movement.

TECHNICAL SKILLS

Programming: Python, C++, CUDA, SystemVerilog

ML / AI: PyTorch, generative models, recursive reasoning, model compression, quantization

Embodied AI / Simulation: RLBench, CoppeliaSim, Genesis

AI Systems: TTNN, NPU profiling, CUDA kernel optimization, memory/layout optimization

Research: Model reproduction, controlled experiments, ablation studies, performance analysis